

PROGRAM ASSESSMENT REPORT

v2024.1

Tracking data

CIM code	
Program name	Master of Data Science Coursera
year	AY 24/25
Program assessment coordinator	Yuhan Ding
Assessment plan year	AY 24/25
No assessment this year per plan	☐
LO(s) assessed	For LO 1 and LO 2, see the description below.
Presented at the program authority meeting date: <i>Please attach minutes</i>	

As outlined in the AY23-24 assessment document, we will evaluate the following program goals for AY 24-25:

LO1: Students will be able to choose the appropriate statistical analysis, data mining, and data-intensive computing methods for the given problem.

LO2: Students will be able to prepare and visualize data for specific problems.

As per the assessment plan, CS 577 Deep Learning (Spring A 25) will assess Learning Outcome 1, while CSP 571 Data Preparation and Analysis (Fall B 24) will assess Learning Outcome 2.

Methodology

For **LO 1**, students will be able to choose the appropriate statistical analysis, data mining, and data-intensive computing methods for a given problem.

- Learning objective one was evaluated through Problem 4 on the summative course assessment administered to all students enrolled in CS 577 Deep Learning during the Spring A 2025 semester.
- Out of 41 students who took the exam, all were enrolled in the Master of Data Science Coursera program.
- The questions and rubric are included. See the next page.

- Assessment question

4. A CNN has the architecture given in the following table. For each layer, calculate the number of parameters and the size of corresponding feature maps. The notation follows the convention as below:

- CONV- K - N - L denotes a convolutional layer with N filters, each of the size $K \times K$, SAME¹ padding and stride = L .
- POOL- K - L indicates a pooling layer with $K \times K$ filter and stride = L .
- FC- N stands for a fully-connected layer with N neurons.

Layer	Feature map shape	Number of parameters
INPUT	$32 \times 32 \times 3$	0
CONV-5-16-2	(1)	(2)
POOL-2-2	(3)	(4)
CONV-3-32-1	(5)	(6)
POOL-2-2	(7)	(8)
Flatten	/	/
FC-10	(9)	(10)

This question is worth 10 points, with one point assigned to each blank.

- Rubrics

0 – 5 points	Does not meet expectations
6 – 8 points	Meets expectations
9– 10 points	Proficient

This problem assesses students' understanding of CNN architecture by calculating the feature map dimensions and the number of trainable parameters at each layer, demonstrating knowledge of how convolution, pooling, and fully connected operations affect the model's structure and complexity.

Students who got 5 points or fewer do not meet the learning objectives. Students may have a limited understanding of CNN layer operations and parameter calculations, suggesting a need for additional practice with network architecture analysis.

Students who got 6 to 8 points demonstrate a general understanding of CNN architecture, with mostly correct calculations of feature map dimensions or parameters. Minor errors or omissions suggest a partial knowledge of layer operations and their impact on model complexity.

Students who received 9 or 10 points demonstrate a clear understanding of CNN architecture by accurately computing both feature map dimensions and parameter counts for all layers. The response demonstrates a strong sense of how convolutional, pooling, and fully connected layers shape the model's structure.

Results

Artifact samples

- The table below displays the number of students who received each specific point total on Problem 1.

Points (out of 12)	Number of students
0	2
3	1
4	1
5	4
6	3
8	1
9	1
10	28

- The table below shows the number and percentage of students at each proficiency level.

Points	Assessments Categories	Number of students	Percentage of students in each category.
0 – 5 points	Does not meet expectations	8	19.51 %
6 – 8 points	Meets expectations	4	9.76%
9 – 10 points	Proficient	29	70.73%

- Proficient Category
Artifact 1: The student obtains complete credit

Input: $32 \times 32 \times 3$ feature map ; total parameters = 0

CONV-5-16-2: $16 \times 16 \times 16$ feature map with each filter $5 \times 5 \times 3 + 1$ bias = 76 parameters; $16 * 76$ = total parameters = 1,216

POOL2-2: $8 \times 8 \times 16$ feature map; total parameters = 0 (no trainable parameters)

CONV-3-32-1: $8 \times 8 \times 32$ feature map with each filter $3 \times 3 \times 16 + 1$ bias = 145 parameters; $32 * 145$ = total parameters = 4,640

POOL2-2: $4 \times 4 \times 32$ feature map; total parameters = 0 (no trainable parameters)

Flatten: $4 \times 4 \times 32 = 512$ feature map; total parameters = 0 (no trainable parameters)

FC-10: 10 feature map with 512 inputs + 1 bias = $513 * 10$ = total parameters = 5,130

Total network parameters = 10,986

Artifact 2: The student obtains 9 out of 10 points.

1) convolutional layer 5-16-2 feature map shape

5x5 filter size

16 filters

stride of 2

output dimensions = $\text{inputDimension} / \text{stride} = 32 / 2 = 16$ by 16

16 filters for a depth of 16

16 x 16 x 16

2) convolutional layer 5-16-2 parameters

filter size of 5x5 with 3 input channels:

$5 \times 5 \times 3 = 75$

one bias channel * 16 filters = $16 \times 76 = \mathbf{1216 \text{ parameters}}$

3) pooling layer 2-2 feature map shape

previous dimensions / stride = 8x8 with the same depth

8x8x16

4) pooling layer 2-2 parameters

0 parameters for a pooling layer

5) CONV-3-32-1 feature map shape

stride of 1 results in no change in the spatial dimensions from the previous layer but now with 32 filters:

8x8x32

6) CONV-3-32-1 parameters

3x3 filter with 16 channels = $3 * 3 * 16 = 144$

plus 1 bias filter yields 32 filters * 145 parameters = **4640 parameters**

7) POOL-2-2 feature map shape

a repeat of the pooling layer from earlier with a 2x2 filter and stride of 2:

$\text{previousDimension} / \text{stride} = 8/2 = 4$

depth remains the same

4x4x32

8) POOL-2-2 parameters

0, no parameters for a pooling layer

9) FC-10 feature map shape

this layer calls for a fully connected layer with 10 neurons for a 10 dimensional vector output

feature map shape = 10

10) FC-10 parameters

10 neurons connected to 4x4x32 inputs from the previous layer:

$10 * 4 * 4 * 32 = 10 * 512 = 5120 + 10 \text{ bias parameters for each neuron} = \mathbf{5130 \text{ parameters}}$

- Meets expectations Category

Artifact 3: The student obtains 8 out of 10 points.

Final Feature Map Shapes:

- CONV-5-16-2 Layer 1: (16, 16, 16)
- POOL-2-2 Layer 2: (8, 8, 16)
- CONV-3-32-1 Layer 3: (8, 8, 32)
- POOL-2-2 Layer 4: (4, 4, 32)
- Flatten Layer 5: (512,)
- FC-10 Layer 6: (10,)

Final Number of Parameters:

- Layer 1: 1216
- Layer 2: 0
- Layer 3: 4640
- Layer 4: 0
- Layer 5: 0
- Layer 6: 5130

Artifact 4: The student earns 6 out of 10 points.

- 1) $16 \times 16 \times 16$ (new size = $(\text{original} - (\text{filter} - 1) + 2 * \text{padding}) / \text{stride} + 1$
Original=32, filter=5, padding=1 (same), stride=2)
- 2) 400 parameters ($16 \times 5 \times 5$)
- 3) $8 \times 8 \times 16$ (dimensions cut in half by 2×2 max pool)
- 4) There are no parameters associated with a pooling layer
- 5) $8 \times 8 \times 32$ (same padding with a stride of one means no size change; channels is equal to the number of filters)
- 6) 288 parameters ($32 * 3 * 3$)
- 7) $4 \times 4 \times 32$ (dimensions cut in half by 2×2 max pool)
- 8) There are no parameters associated with a pooling layer
- 9) 512×1 , equal to the previous layer's dimensions multiplied out
- 10) 5130 parameters, 512 weights plus a bias per neuron

- Does not meet expectations category

Artifact 6: The student obtains 5 out of 10 points

1. $13 \times 13 \times 16$
2. 1216
3. $6 \times 6 \times 16$
4. 0
5. $4 \times 4 \times 32$
6. 4640
7. $2 \times 2 \times 32$
8. 0
9. 1×10
10. 1290

Artifact 7: The student obtains 4 out of 10 points

Layer parameters	Feature map shape	Number of
INPUT 0	32x32x3	
CONV-5-16-2 1,216	32x32x16	
POOL-2-2 0	16x16x16	
CONV-3-32-1 4,640	16x16x32	
POOL-2-2 0	8x8x32	
Flatten 0	2,048	
FC-10 20,490	10	

Artifact 8: The student got 3 out of 10 points

CONV-5-16-2 denotes a convolutional layer with 16 filters, each of size 5x5, SAME1 and stride = 2. Output size 14x14x16. 1216 parameters.

POOL-2-2 indicates a pooling layer with 2x2 filter and stride 2. Output size 7x7x16.

CONV-3-32-1 denotes a convolutional layer with 32 filters, each of size 3x3, SAME1 and stride 1. Output size 7x7x32. 4640 parameters.

POOL-2-2 indicates a pooling layer with 2x2 filter and stride 2. Output size 3x3x32.

Flatten size is 288.

FC-10 stands for a fully connected layer with 10 neurons. 2890 parameters.

Artifact 9: The student got 0 out of 10 points

Size = 5

stride = 2

First Convolution CONV-5-16-2 :

output = (input-size/stride) = $(32-5/2) + 1 = 27/2 + 1 = 13$

Feature map shape = 13 X 13 X 3

No of parameters per filter : $5 \times 5 \times 3 = 75$

Number of parameters = $75 \times 16 = 1200$

POOL 2-2 : Hight and width = $(13-2)/2 + 1$ and $(13-2)/2 + 1 = (11/2) + 1$ and $(11/2) + 1 = 5.5 + 1$ and $5.5 + 1 = 6.5$ and 6.5 and rounded value is 6

feature map = 6 X 6 X 16

parameters = 1200

CONV-3-32-1 :

Feature map = 30X30X32

Number parameters = $3 \times 3 \times 32 = 288$

POOL-2-2 : 15X15X32

Number of parameters = 288

Pattern size = $15 \times 15 \times 32 = 7200$

FC-10 = $7200 \times 10 = 72000$

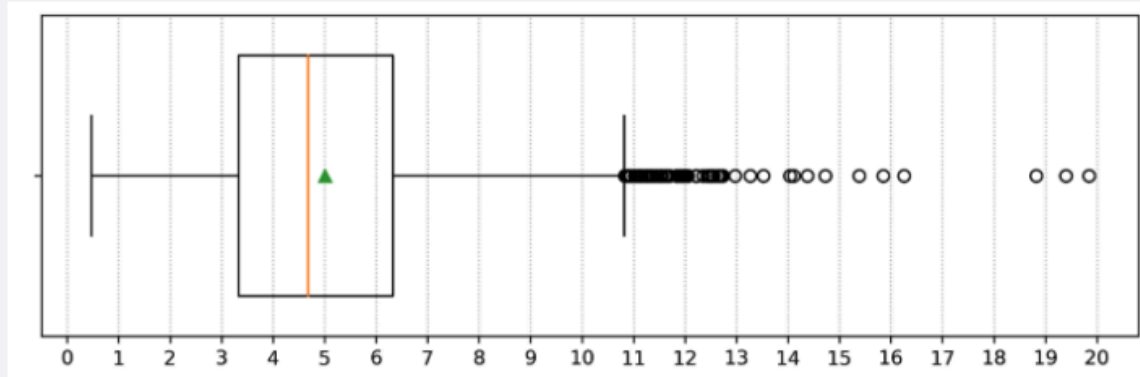
- The data indicate that 80.49% of students meet the learning objective, with 70.73% achieving proficiency.

Methodology

For **LO 2**, students will be able to prepare and visualize data for specific problems.

- Learning objective two was evaluated through Problem 1 on the summative course assessment administered to all students enrolled in CSP 571 Data Preparation and Analysis during the Fall B 2024 semester.
- Out of 47 students who took the exam, all were enrolled in the Master of Data Science Coursera program.
- Assessment question

Based on the following boxplot, what is the 50th percentile value?



Options

☐ 2.9979

⊗ **Incorrect**

Incorrect. Which chart element does represent the 50th percentile?

☐ 3.3284

⊗ **Incorrect**

Incorrect. Which chart element does represent the 50th percentile?

☒ 4.6784

✓ **Correct**

Correct. The middle orange line inside the central box marks the 50th percentile. Therefore, this answer best represents the 50th percentile.

☐ 5.0064

⊗ **Incorrect**

Incorrect. Which chart element does represent the 50th percentile?

Results

The table below shows the number and percentage of students at each proficiency level.

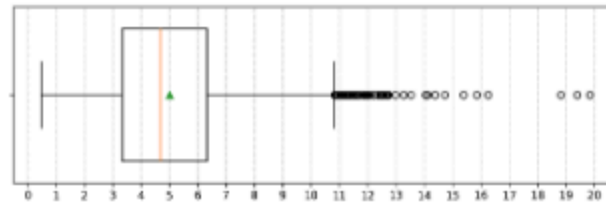
Rubric Category	Assessments Categories	Number of students	Percentage of students in each category.
Incorrect	Does not meet expectations	9	19.15%
Correct	Meets expectations	38	81.85%

The data shows that most students meet the learning objective. These results indicate that students have satisfactorily developed these data preparation and visualization skills. We have included two group artifacts for categories: "Meets Expectations" and "Does Not Meet Expectations."

- Meet Expectations Category

Artifact 1:

1. Based on the following boxplot, what is the 50th percentile value?



- ☐ 2.9979
- ☐ 5.0064
- ☐ 3.3284
- ☒ 4.6784

✓ **Correct**

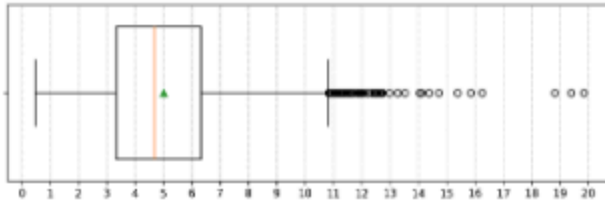
Correct. The middle orange line inside the central box marks the 50th percentile. Therefore, this answer best represents the 50th percentile.

- Does not meet expectations category

Artifact 2:

1. Based on the following boxplot, what is the 50th percentile value?

0 / 1 point



- ☐ 2.9979
- ☐ 4.6784
- ☐ 3.3284
- ☒ 5.0064

⊗ Incorrect

Incorrect. Which chart element does represent the 50th percentile?

Even though nine students answered this question incorrectly, they all picked the same wrong answer: 5.0064.

Continuous Improvement, CI, plans

The assessment data highlights essential insights into our students' performance on specific learning objectives. Here's an analysis of the findings and our improvement plans:

Learning Objective 1: 80.49% of our students successfully achieved the learning objective focused on developing the appropriate model for the given problem, indicating a strong ability to apply theoretical concepts to specific issues.

Plan for Improvement: Although 80.49% of Coursera students met the learning objective, this rate is lower than that of the regular program in last year's assessment. The data suggest that additional instructional strategies are needed to strengthen students' understanding of convolutional neural network (CNN) architecture and parameter analysis. To enhance student learning outcomes, the following actions are proposed:

1. **Enhanced instructional materials:** Develop additional step-by-step examples and annotated explanations to clarify the relationship between layer configuration, feature map dimensions, and parameter calculations.
2. **Interactive practice opportunities:** Integrate formative assessments and self-paced exercises that allow students to apply key formulas and receive automated feedback.
3. **Supplemental learning support:** Offer optional review sessions or discussion forums focused on CNN layer operations to address common misconceptions.
4. Provide more detailed feedback on assignments to help students identify and correct calculation or interpretation errors.
5. **Continuous monitoring:** Evaluate the impact of these interventions in subsequent offerings by comparing performance data and qualitative student feedback.

Learning Objective 2: 81.85% of students have successfully met the learning objective, indicating that most students are competent in data preparation and visualization. We aim to maintain a high level of achievement while deepening student proficiency, introducing advanced techniques, and promoting creative, critical thinking in data handling and visualization.

Plan for Improvement: Although **81.85% of students met the learning objective**, this rate is lower than that of the regular program in last year's assessment. The data suggest that additional instructional strategies are needed to strengthen students' ability to interpret boxplots and identify key summary statistics such as the median (50th percentile). To enhance student learning outcomes, the following actions are proposed:

1. Incorporate more visual examples and practice questions focused on interpreting boxplots and percentiles.
2. Provide short video demonstrations explaining how to read boxplots step by step.
3. Add interactive exercises where students match graphical elements to statistical terms (median, quartiles, outliers).
4. Reinforce the concept through in-class discussions and formative quizzes that emphasize visual reasoning.